

RRC-Raptor-V4H

- Porting Guide

V0.1.0

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i. VERSION LIST

Description	Version	Date	Author
Porting Guide	V0.0.1	2024/09/12	kevinlin
Modify 3.5(Page14) 、 4.1(Page16~Page22) 、 4.4(Page25)	V0.0.2	2024/10/28	kevinlin
Modify 3.2(Page10) 、 4.1.3(Page19) , File attribute changes	V0.0.3	2024/12/19	kevinlin
Adjust BSP 3p28 hyperlink Cover range: - Section 3.4 - Section 4.1.2 - Section 5	V0.0.4	2026/03/06	Felix Hsu
- Modify content for readability - Rename Section title - Merge part of the section - Fix typo and out-of-date content Cover range: ALL	V0.1.0	2026/03/06	Felix Hsu

ii. HASHTAG

#V4H

#u-boot

#kernel

#BSP

#ubuntu

1. Overview

1.1. Target

This guide introduces RRC Raptor, a product extended from R-Car V4H, build method of Linux kernel, U-boot, root filesystem, and how to change the IPL (initial program loader).

1.2. Scope

This manual is only used for RRC Raptor carrier board with RCar-V4H SoM.

2. Host System Requirements

Linux Host PC is used as building BSP and provides TFTP and NFS server for booting. The minimum disk space required is about 50 GB. It is recommended that at least 120 GB is provided, which is enough to compile all backends together. The recommended minimum OS version is Ubuntu 20.04 LTS (64 bit only) or later. Reference host PC specifications are listed as follows:

Reference Linux Host PC software:

OS name : Ubuntu 20.04.4 LTS
Codename : focal
Kernel : 5.15.0-43-generic

Reference Linux Host PC hardware:

Processor : Intel i9-11900K @ 2.50GHz
Memory : 64GiB
Block device : 7.3TB Disk

3. Building BSP

This chapter introduces the procedure about how to build BSP by using Yocto project. Please execute the following steps on Linux host PC to complete the build.

For Yocto project, please refer to the following URL for details:

<https://docs.yoctoproject.org/current/brief-yoctoprojectqs/index.html>

3.1. Prerequisite

Step 1: Install all building prerequisites on Linux Host PC

Building essential packages for Linux Host PC are listed below:

```
$ sudo apt-get install gawk wget git-core diffstat unzip texinfo gcc-multilib  
build-essential chrpath socat cpio python python3 python3-pip python3-pexpect  
xz-utils debianutils iputils-ping python3-git python3-jinja2 libegl1-mesa  
libSDL1.2-dev pylint3 xterm libarchive-zip-perl rsync curl zstd lz4 libssl-dev
```

Step 2: Set up git

This step may not be needed if git has been set up

```
$ git config --global user.name "Your Name"  
$ git config --global user.email "Your Email"  
$ git config --list
```

3.2. Using BSP downloader

Step 3: Fetch the BSP downloader

To simplify the procedure of building BSP, a BSP building script is provided

```
$ mkdir RRC-Raptor-V4H  
$ cd RRC-Raptor-V4H  
$ git clone https://github.com/RetronixTechInc/rcar-bsp -b v4h-raptor
```

Step 4: Use the downloader to automatically complete the building procedure

```
$ cd rcar-bsp  
$ ./build_yocto_v3p40.sh
```

Notice: Yocto project gather all building information and intermediate files under the path “\${WORK}/build-*”. To prevent making the build become a mess, any of different projects sharing the same path name shall be removed to allow the future builds to start from a clean environment.

3.2.1. Confirm the build information

Step 4-1 (optional)

Yocto project shows some key information during building used to ensure the build uses the correct materials and configuration. The following details are provided for confirmation:

```
BB_VERSION      = 1.46.0
BUILD_SYS       = x86_64-linux
TARGET_SYS      = aarch64-poky-linux
MACHINE         = raptor
DISTRO          = poky
DISTRO_VERSION  = 3.1.11
SOC_FAMILY      = rcar:rcar-v4x:r8a779g0
```

META

```
meta-yocto-bsp   = tmp: 74b22db6879b388d700f61e08cb3f239cf940d18
meta-rcar-adas   = tmp: 89ca1415e3598789e9004363f9a38c1b3f41912d
meta-rtx         = v4h-raptor/meta-rtx:${AUTOREV}
meta-networking  = tmp: 814eec96c2a29172da57a425a3609f8b6fcc6afe
```

3.3. File Deployment

After successfully building, the deployed files will be created under the specific path “\${WORK}/build-raptor-adas/tmp/deploy/images/raptor/”

Image	← kernel file
r8a779g0-raptor.dtb	← kernel device tree
r8a779g0-raptor-overlay-pcie1x4.dtb	← kernel device tree overlay
r8a779g0-raptor-overlay-dsi0-7.dtb	← kernel device tree overlay
r8a779g0-raptor-overlay-dsi0-10.dtb	← kernel device tree overlay
r8a779g0-raptor-overlay-dsi1-7.dtb	← kernel device tree overlay
r8a779g0-raptor-overlay-dsi1-10.dtb	← kernel device tree overlay
rcar-image-adas-v4h-\${DATE}.tar.bz2	← file system
u-boot-elf-raptor.srec	← u-boot file

4. Install BSP to the machine

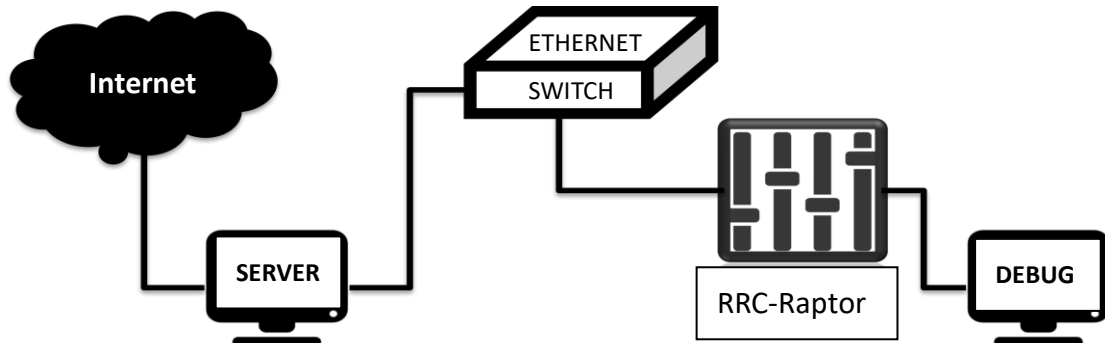


Figure 1. Recommended Environment

By default, RRC-Raptor DIP Switch SW26 – SW32 state should be as follow:

Value from left to right represents the ON/OFF state from 1 to 8

SW26	SW35	SW31	SW27	SW30
1111 1111	1111 1111	1111 1111	1111 1111	1111 1111
SW28	SW36	SW33	SW29	SW32
0100 1011	0111 1110	1111 0100	1111 0111	1111 1111

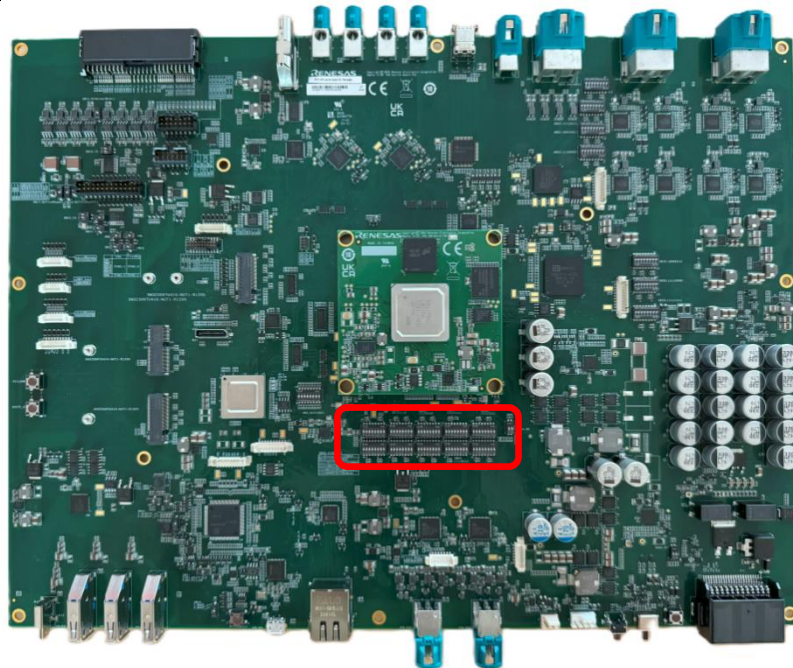


Figure 2. DIP switch of Boot mode

4.1. Flashing Boot Loaders

This Section introduces how to replace the u-boot image and flash IPL to the machine.

Step 1: Change the DIP switch to Download mode

SW26	SW35	SW31	SW27	SW30
1111 1111	1111 1111	1111 1111	1111 1111	1111 1111
SW28	SW36	SW33	SW29	SW32
0100 0000	0111 1110	1111 0100	1111 0111	1111 1111

Step 2: Download IPL images

Please refer to <https://github.com/RetronixTechInc/rcar-bsp> or check the link below.

URL: [RRC-V4H-Raptor-IPL.zip](#)

Step 3: Extract RRC-V4H-Raptor-IPL.zip

Extract RRC-V4H-Raptor-IPL.zip and you'll see files listed below:

```

| bl31.srec
| bootparam_sa0.srec
| cert_header_sa9.srec
| cr52_loader.srec
| dummy_fw.srec
| dummy_rtos.srec
| FlashWriter_V4H.sh
| ICUMX_Flash_writer_SCIF_DUMMY_CERT_EB203000_V4H.mot
| icumx_loader.srec
| tee.srec
└── u-boot-elf.srec ← u-boot file

```

Step 4: Replace the U-boot image

Please refer to section [4.1.1](#) below.

Step 5: Flash the IPL images

Please refer to section [4.1.2](#) below.

Step 6: Power off RRC Raptor

Step 7: Change the DIP switch back to Boot mode

SW26	SW35	SW31	SW27	SW30
1111 1111	1111 1111	1111 1111	1111 1111	1111 1111
SW28	SW36	SW33	SW29	SW32
0100 1011	0111 1110	1111 0100	1111 0111	1111 1111

4.1.1. Replacing the U-boot image

Replace the **u-boot-elf.srec** under the extracted RRC-V4H-Raptor-IPL folder by **u-boot-elf-raptor.srec** deployed from section [File Deployment](#). Other files remain unchanged.

4.1.2. Flashing IPL

Before flashing IPL, please ensure the following two things:

- DIP switch is under download mode
- RRC Raptor is power on

Step 5-1: Connect USB port

Connect USB port on Ubuntu Host PC to Carried Board (CN23) through USB cable for the console

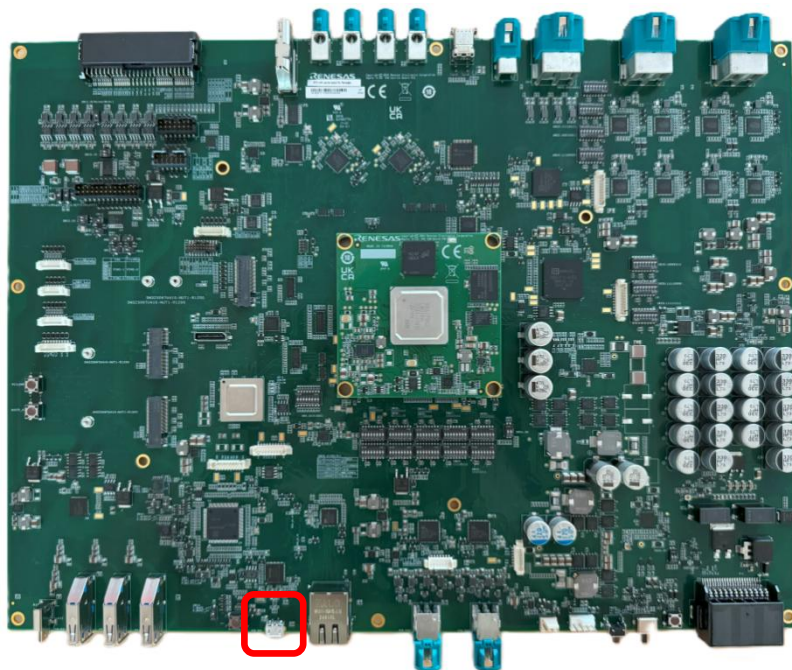


Figure 3. CN23 – USB port for debugging

Step 5-2: Connect RRC Raptor by using terminal software

Open a terminal software like PuTTY on Ubuntu Host PC. Configure the serial line name and baud rate as below:

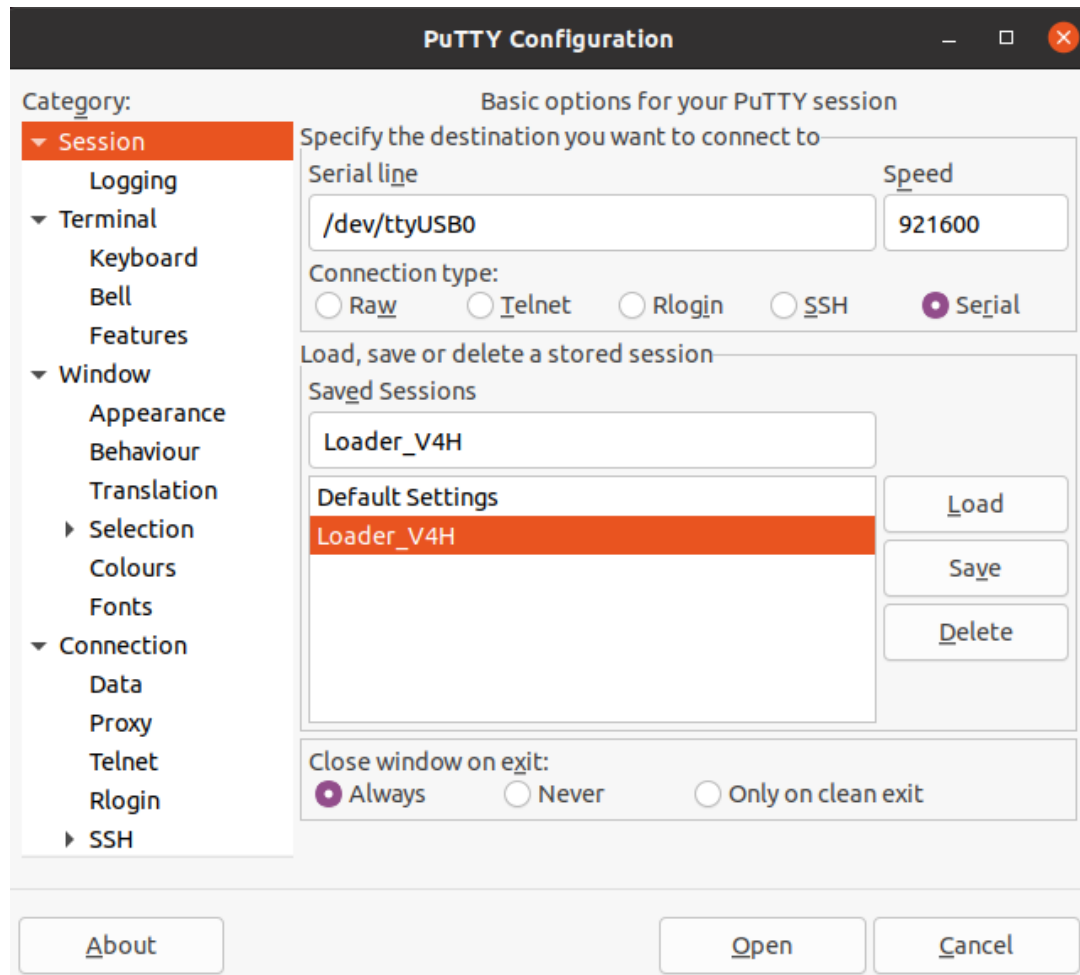


Figure 4. PuTTY setting

Step 5-3: Execute FlashWriter tool on Host PC

Open the other terminal and execute “FlashWriter_V4H.sh” on host PC to download firmware to QSPI Flash and eMMC

```
$ cd RRC-V4H-Raptor-IPL  
$ sudo ./FlashWriter_V4H.sh -a
```

FlashWriter will automatically return. Once it is finished, the firmware download is completed.

4.2. TFTP Server

This section introduces how to install TFTP server on Host PC for RRC Raptor.

Please confirm the Host PC is connected to the internet. If there are any errors, try [*apt update -y*](#) to update the application repository database.

Step 1: Install tftpd-hpa

```
$ sudo apt install tftpd-hpa -y
```

Step 2: Modify "TFTP_OPTIONS" in tftpd-hpa to give R/W permission to TFTP folder

```
$ sudo vi /etc/default/tftpd-hpa
```

```
# /etc/default/tftpd-hpa

TFTP_USERNAME="tftp"
TFTP_DIRECTORY="/var/lib/tftpboot"
TFTP_ADDRESS=":69"
#TFTP_OPTIONS="--secure"
TFTP_OPTIONS="--secure --create"
```

Step 3: Restart the tftpd-hpa service

```
$ sudo service tftpd-hpa restart
```

4.3. NFS Server

This section introduces how to install NFS server on host PC for RRC Raptor.

Step 1: Install NFS kernel server package and rpcbind

```
$ sudo apt install nfs-kernel-server  
$ sudo apt install rpcbind
```

Step 2: Configure the NFS Server

```
$ sudo vi /etc/exports
```

```
/export/rfs *(rw,async,no_root_squash)
```

Step 3: Notify the NFS server after making any changes to the export file

```
$ exportfs -a
```

Step 4: Restart the NFS Service

```
$ sudo service nfs-kernel-server restart
```

4.4. Starting of U-Boot and Linux

[Host PC]

Step 1: Prepare root filesystem, kernel image and kernel device tree

Those files are listed in section [File Deployment](#). Please copy the kernel image and device tree to TFTP root directory.

```
$ mkdir -p /var/lib/tftpboot/root/  
$ cp Image /var/lib/tftpboot/root/  
$ cp r8a779g0-raptor.dtb /var/lib/tftpboot  
$ cp r8a779g0-raptor-overlay-pcie1x4.dtb /var/lib/tftpboot  
$ cp r8a779g0-raptor-overlay-dsi0-7.dtb /var/lib/tftpboot  
$ cp r8a779g0-raptor-overlay-dsi0-10.dtb /var/lib/tftpboot  
$ cp r8a779g0-raptor-overlay-dsi1-7.dtb /var/lib/tftpboot  
$ cp r8a779g0-raptor-overlay-dsi1-10.dtb /var/lib/tftpboot
```

Copy root filesystem rcar-image-adas-v4h- $\{DATE\}$.tar.bz2 to root directory of NFS server, and extract it as below

```
$ cp rcar-image-adas-v4h- $\{DATE\}$ .tar.bz2 /export/rfs  
$ cd /export/rfs  
$ tar xvf rcar-image-adas-v4h- $\{DATE\}$ .tar.bz2
```

[RRC-Raptor]

The following steps are necessary only the first time entering the system. Once those boot parameters are configured, the system can boot up directly next time.

Step 1: Boot up and stop autobooting

Power on RRC Raptor. When the system executes the u-boot image, the console screen displays "Hit any key to stop autoboot: *". Hit any key to enter command mode

```
Hit any key to stop autoboot: 0  
=>
```

Step 2: Reset DEFAULT environment variables

```
=> env default -a
```

Step 3: Set U-Boot environment variables

The command indicates to mount the root file system through the **NFS server with read/write options, the IP is 192.168.0.1** (default sample), the root directory is /export/rfs, and the **client IP is 192.168.0.20** (default sample). (IP value depends on domain settings)

xx:xx:xx:xx:xx:xx is MAC address of AVB0 port

```
=> setenv -f ethaddr xx:xx:xx:xx:xx:xx
=> setenv ipaddr xx.xx.xx.xx
=> setenv serverip xx.xx.xx.xx
```

Step 4: Change the bootargs by U-Boot

The default *bootcmd* runs *bootcmd_emmc*. It needs to be modified to *bootcmd_nfs* to let the system connect TFTP and NFS server

```
=> setenv bootcmd 'run bootcmd_nfs'
```

Step 5: Save environment variables to persistent storage

```
=> saveenv
```

Step 6: Restart the system

```
=> reset
```

5. APPENDIX

RETROPIX Source Code Link :

<https://github.com/RetronixTechInc/rcar-bsp>

README.md at v4h-raptor :

[README.md](#)

Download IPL images :

[RRC-V4H-Raptor-IPL.zip](#)